

Solutions of Tutorial #3

Topics covered:

- Closed sets and limit points

Problem 1

Let A, B , and A_α denote subsets of a space X . Prove the following:

- (a) If $A \subset B$, then $\bar{A} \subset \bar{B}$.
- (b) $\overline{A \cup B} = \bar{A} \cup \bar{B}$.
- (c) $\overline{\bigcup A_\alpha} \supset \bigcup \bar{A}_\alpha$; give an example where equality fails.

Solution: (a) Suppose that $A \subset B$ and consider any $x \in \bar{A}$. Consider any neighborhood U of x so that U intersects A by Theorem 17.5 part (a). Hence there is a point $y \in U \cap A$ so that $y \in U$ and $y \in A$. But then clearly $y \in B$ also since $A \subset B$. Therefore $y \in U \cap B$ so that U intersects B . Since U was an arbitrary neighborhood of x , this shows that $x \in \bar{B}$, again by Theorem 17.5 part (a). This of course shows that $\bar{A} \subset \bar{B}$ as desired since x was arbitrary.

(b) ($\subset$) We show this by contrapositive. So suppose that $x \notin \bar{A} \cup \bar{B}$. Then clearly $x \notin \bar{A}$ and $x \notin \bar{B}$. Thus, by Theorem 17.5 part (a), there is an open set U_A such that U_A does not intersect A , and likewise an open U_B that does not intersect B . Let $U = U_A \cap U_B$, which is clearly open since U_A and U_B are. We also note that U contains x since both U_A and U_B do. Then it must be that U does not intersect A since, if it did, then U_A would also intersect A since $U \subset U_A$. Similarly, U cannot intersect B . Thus, for all $y \in U$, $y \notin A$ and $y \notin B$. This is logically equivalent to saying that there is no $y \in U$ where $y \in A$ or $y \in B$, therefore there is no $y \in U$ where $y \in A \cup B$. Hence U and $A \cup B$ do not intersect. Since U is open and contains x , this shows that $x \notin \overline{A \cup B}$, again by Theorem 17.5 part (a). Therefore, by contrapositive, $x \in \overline{A \cup B}$ implies that $x \in \bar{A} \cup \bar{B}$ so that $\overline{A \cup B} \subset \bar{A} \cup \bar{B}$.

($\supset$) Consider any $x \in \bar{A} \cup \bar{B}$ and any neighborhood U of x . If $x \in \bar{A}$ then U intersects A by Theorem 17.5 part (a). Hence there is a $y \in U \cap A$ so that $y \in U$ and $y \in A$. Then clearly $y \in A \cup B$ so that y is also in $U \cap (A \cup B)$. Hence U intersects $A \cup B$. An analogous argument shows that this is also true if $x \in \bar{B}$ instead. Since U was an arbitrary neighborhood, this shows that $x \in \overline{A \cup B}$ by Theorem 17.5 part (a). Hence $\bar{A} \cup \bar{B} \subset \overline{A \cup B}$ since x was arbitrary.

(c) Consider any $x \in \bigcup \bar{A}_\alpha$ so that there is a particular β where $x \in \bar{A}_\beta$. Suppose that U is any open set containing x so that U intersects A_β by Theorem 17.5 part (a) since $x \in \bar{A}_\beta$. Then clearly U also intersects $\bigcup A_\alpha$ since $A_\beta \subset \bigcup A_\alpha$. Since U was an arbitrary open set containing x , this shows that $x \in \overline{\bigcup A_\alpha}$ by Theorem 17.5 part (a). This shows that $\bigcup \bar{A}_\alpha \subset \overline{\bigcup A_\alpha}$ since x was arbitrary, which is of course the desired result.

As an example where equality fails, consider the standard topology on $\mathbb{R}$ and the sets $A_n = (1/n, 2]$ for $n \in \mathbb{Z}_+$. It is then trivial to show that $\bigcup A_n = (0, 2]$ so that clearly 0 is a limit point of $\bigcup A_n$, and hence $0 \in \overline{\bigcup A_n}$. However, for any $n \in \mathbb{Z}_+$, the open interval $(-1, 1/n)$ is clearly an open set containing 0 that is disjoint from $(1/n, 2] = A_n$. This shows that $0 \notin \bar{A}_n$ for every $n \in \mathbb{Z}_+$ by Theorem 17.5 part (a), from which it follows that $0 \notin \bigcup \bar{A}_n$. Hence $\bigcup \bar{A}_n$ is not a subset of $\overline{\bigcup A_n}$ and thus $\bigcup \bar{A}_n \neq \overline{\bigcup A_n}$.

Problem 2

Let A, B , and A_α denote subsets of a space X . Determine whether the following equations hold; if an equality fails, determine whether one of the inclusions $\supset$ or $\subset$ holds.

(a) $\overline{A \cap B} = \bar{A} \cap \bar{B}$.

(b) $\overline{\bigcap A_\alpha} = \bigcap \bar{A}_\alpha$.

(c) $\overline{A - B} = \bar{A} - \bar{B}$.

Solution: (a) We claim that $\overline{A \cap B} \subset \bar{A} \cap \bar{B}$ but equality is not always true. Indeed, consider any $x \in \overline{A \cap B}$ and any open set U containing x . Then by Theorem 17.5 part (a), U intersects $A \cap B$, from which it immediately follows that U intersects both A and B . However, since U was an arbitrary neighborhood of x , it follows from Theorem 17.5 part (a) again that x is in both $\bar{A}$ and $\bar{B}$. Hence $x \in \bar{A} \cap \bar{B}$, which shows that $\overline{A \cap B} \subset \bar{A} \cap \bar{B}$ since x was arbitrary.

Now consider the standard topology on $\mathbb{R}$ with $A = [-1, 0)$ and $B = (0, 1]$. As these are clearly disjoint, we have that $A \cap B = \emptyset$ so that $\overline{A \cap B} = \emptyset$ also. However, since we also clearly have that $\bar{A} = [-1, 0]$ and $\bar{B} = [0, 1]$, it follows that $\bar{A} \cap \bar{B} = \{0\}$. Thus clearly $\overline{A \cap B} = \emptyset \neq \{0\} = \bar{A} \cap \bar{B}$ as desired.

(b) We again claim that $\overline{\bigcap A_\alpha} \subset \bigcap \bar{A}_\alpha$ but that equality is not generally true. Indeed, consider any $x \in \overline{\bigcap A_\alpha}$ and any open set U of x . Then, by Theorem 17.5 part (a), U intersects $\bigcap A_\alpha$ so that, for any particular A_β , U intersects A_β . This shows that $x \in \bar{A}_\beta$ by Theorem 17.5 part (a) so that $x \in \bar{A}_\alpha$ for every α since β was arbitrary. Hence $x \in \bigcap \bar{A}_\alpha$, which shows that $\overline{\bigcap A_\alpha} \subset \bigcap \bar{A}_\alpha$ since x was arbitrary. As in part (a), equality fails if we have $A_1 = [-1, 0)$ and $A_2 = (0, 1]$ in the standard topology on $\mathbb{R}$. By the same argument as in part (a) it follows that $\bigcap_{n=1}^2 A_n = \emptyset \neq \{0\} = \bigcap_{n=1}^2 \bar{A}_n$.

(c) Here we claim that $\overline{A - B} \supset \bar{A} - \bar{B}$ but that the converse does not always hold. Consider any $x \in \bar{A} - \bar{B}$ and any open set U containing x . Then $x \in \bar{A}$ so that every open set

containing x intersects A by Theorem 17.5 part (a). Also $x \notin \bar{B}$ so that there is an open set V containing x that does not intersect B , also by Theorem 17.5 part (a). Let $W = U \cap V$ so that W contains x since both $x \in U$ and $x \in V$. Now, since W is also an open set containing x , W intersects A so that there is a $y \in W$ where also $y \in A$. It also cannot be that $y \in B$ since we have $y \in W \subset V$ so that then V would intersect B . Therefore $y \in A - B$. Also we have $y \in W \subset U$ so that also $y \in U$. Hence U intersects $A - B$, which shows that $x \in \overline{A - B}$ by Theorem 17.5 part (a) since U was an arbitrary neighborhood of x . Therefore $\overline{A - B} \supset \bar{A} - \bar{B}$ as desired since x was arbitrary.

As a counterexample to equality, consider the standard topology on $\mathbb{R}$ with $A = [0, 2]$ and $B = (1, 3]$. Then clearly $\bar{A} = A = [0, 2]$ and $\bar{B} = [1, 3]$, from which it is easily shown that $\bar{A} - \bar{B} = [0, 1)$. But we also have $A - B = [0, 1]$ so that obviously $\overline{A - B} = [0, 1]$ as well. Therefore $\overline{A - B} = [0, 1] \neq [0, 1) = \bar{A} - \bar{B}$ as desired.

Problem 3

Let $A \subset X$ and $B \subset Y$. Show that in the space $X \times Y$,

$$\overline{A \times B} = \bar{A} \times \bar{B}.$$

Solution: ($\subset$) Consider $(x, y) \in \overline{A \times B}$. Also suppose that U and V are any open sets in X and Y , respectively, that contain x and y , respectively. Then $U \times V$ is a basis element of the product topology on $X \times Y$, by definition, that contains (x, y) . It then follows from Theorem 17.5 part (b) that $U \times V$ intersects $A \times B$ and hence there is a point $(w, z) \in U \times V$ where also $(w, z) \in A \times B$. Then $w \in U$ and $w \in A$ so that U intersects A , and hence $x \in \bar{A}$ by Theorem 17.5 part (a) since U was an arbitrary neighborhood of x . An analogous argument shows that $y \in \bar{B}$. Therefore $(x, y) \in \bar{A} \times \bar{B}$ so that $\bar{A} \times \bar{B} \subset \overline{A \times B}$ since x was arbitrary.

($\supset$) Now suppose that (x, y) is any point in $\bar{A} \times \bar{B}$ so that $x \in \bar{A}$ and $y \in \bar{B}$. Suppose also that $U \times V$ is any basis element of $X \times Y$ that contains (x, y) so that by definition U and V are open in X and Y , respectively. Since $x \in \bar{A}$ and U is an open set where $x \in U$, it follows from Theorem 17.5 part (a) that U intersects A . Thus there is $w \in U$ where $w \in A$ as well. An analogous argument shows that V intersects B so that there is a $z \in V$ where also $z \in B$. We therefore have that $(w, z) \in U \times V$ and $(w, z) \in A \times B$ so that $U \times V$ intersects $A \times B$. Since $U \times V$ was an arbitrary basis element containing (x, y) , it follows from Theorem 17.5 part (b) that $(x, y) \in \overline{A \times B}$. This shows that $\bar{A} \times \bar{B} \subset \overline{A \times B}$ since the point (x, y) was arbitrary.

Problem 4

Show that the product of two Hausdorff spaces is Hausdorff.

Solution: Suppose that X and Y are Hausdorff spaces and consider two distinct points (x_1, y_1) and (x_2, y_2) in $X \times Y$. Since these points are distinct, it has to be that $x_1 \neq x_2$

or $y_1 \neq y_2$. In the first case x_1 and x_2 are distinct points of X so that there are disjoint neighborhoods U_1 and U_2 of x_1 and x_2 , respectively. This of course follows from the fact that X is a Hausdorff space. Then we have that $U_1 \times Y$ and $U_2 \times Y$ are both basis elements, and therefore open sets, in the product space $X \times Y$ since Y itself is obviously an open set of Y . Clearly also $(x_1, y_1) \in U_1 \times Y$ and $(x_2, y_2) \in U_2 \times Y$ so that $U_1 \times Y$ is a neighborhood of (x_1, y_1) and $U_2 \times Y$ is a neighborhood of (x_2, y_2) .

Then, for any $(x, y) \in U_1 \times Y$ we have that $x \in U_1$ so that $x \notin U_2$ since they are disjoint. Then it has to be that $(x, y) \notin U_2 \times Y$. This suffices to show that $U_1 \times Y$ and $U_2 \times Y$ are disjoint since (x, y) was arbitrary. Thus $X \times Y$ is a Hausdorff space since the points (x_1, y_1) and (x_2, y_2) were arbitrary. An analogous argument in the case in which $y_1 \neq y_2$ shows the same result.

Problem 5

Show that a subspace of a Hausdorff space is Hausdorff.

Solution: Suppose that X is a Hausdorff space and that Y is a subset of X . Consider any two distinct points y_1 and y_2 in Y so that of course also $y_1, y_2 \in X$. Then there are neighborhoods U_1 and U_2 of y_1 and y_2 , respectively, that are disjoint since X is Hausdorff. Since U_1 is open in X , we have that $V_1 = U_1 \cap Y$ is open in Y by the definition of a subspace topology. Clearly also V_1 contains y_1 since $y_1 \in U_1$ and $y_1 \in Y$. Similarly $V_2 = U_2 \cap Y$ is an open set of Y that contains y_2 . Then, for any $x \in V_1$ clearly $x \in U_1$ so that $x \notin U_2$ since U_1 and U_2 are disjoint. Then $x \notin U_2 \cap Y = V_2$. Since x was arbitrary, this shows that V_1 and V_2 are disjoint, which then shows that Y is a Hausdorff space as desired.

Problem 6

If $A \subset X$, we define the boundary of A by the equation

$$\text{Bd } A = \bar{A} \cap \overline{(X - A)}.$$

- (a) Show that $\text{Int } A$ and $\text{Bd } A$ are disjoint, and $\bar{A} = \text{Int } A \cup \text{Bd } A$.
- (b) Show that $\text{Bd } A = \emptyset \Leftrightarrow A$ is both open and closed.
- (c) Show that U is open $\Leftrightarrow \text{Bd } U = \bar{U} - U$.
- (d) If U is open, is it true that $U = \text{Int}(\bar{U})$? Justify your answer.

Solution: (a) Consider any $x \in \text{Int } A$ so that there is a neighborhood of x that is entirely contained in A . Then, for any $y \in U$, we have that $y \in A$ and hence $y \notin X - A$. This shows that U does not intersect $X - A$, which suffices to show that x is not in the closure of $X - A$ by Theorem 17.5 part (a). Thus x is not in the boundary of A since $\text{Bd } A = \bar{A} \cap \overline{(X - A)}$. This of course shows that $\text{Int } A$ and $\text{Bd } A$ are disjoint since x was arbitrary. To show that

$\bar{A} = \text{Int } A \cup \text{Bd } A$, first consider any $x \in \bar{A}$. If $x \in \text{Int } A$ then clearly $x \in \text{Int } A \cup \text{Bd } A$, so assume that $x \notin \text{Int } A$. Consider any neighborhood U of x . Then it has to be that U is not a subset of A since otherwise x would be in the union of open subsets of A and hence in the interior. It then follows that there is a point $y \in U$ where $y \notin A$ and therefore $y \in X - A$. This shows that U intersects $X - A$ so that x is in the closure of $X - A$ since U was an arbitrary neighborhood. Since also $x \in \bar{A}$, we have that $x \in \bar{A} \cap \overline{(X - A)} = \text{Bd } A$. Hence clearly $x \in \text{Int } A \cup \text{Bd } A$ so that $\bar{A} \subset \text{Int } A \cup \text{Bd } A$ since x was arbitrary.

Now consider any $x \in \text{Int } A \cup \text{Bd } A$. If $x \in \text{Int } A$ then also $x \in \bar{A}$ since we have that $\text{Int } A \subset A \subset \bar{A}$. On the other hand, if $x \in \text{Bd } A = \bar{A} \cap \overline{(X - A)}$ then of course $x \in \bar{A}$. This shows that $\text{Int } A \cup \text{Bd } A \subset \bar{A}$ in either case since x was arbitrary. Since both directions have been shown, it follows that $\bar{A} = \text{Int } A \cup \text{Bd } A$ as desired.

(b) ($\Rightarrow$) First suppose that $\text{Bd } A = \emptyset$. Then by part (a) we have that $\bar{A} = \text{Int } A \cup \text{Bd } A = \text{Int } A \cup \emptyset = \text{Int } A$. Hence $A \subset \bar{A} = \text{Int } A$ so that $A = \text{Int } A$ since it is also always the case that $\text{Int } A \subset A$. This shows that A is open since $\text{Int } A$ is always open. We also have $\bar{A} = \text{Int } A \subset A$ so that $A = \bar{A}$ since it is always also the case that $A \subset \bar{A}$. This of course shows that A is also closed since $\bar{A}$ is always closed.

($\Leftarrow$) Now suppose that A is both open and closed. It then follows that $\bar{A} = A = \text{Int } A$. So consider any $x \in \bar{A}$ so that also $x \in \text{Int } A$. Then there is a neighborhood U of x contained entirely in A . Thus, for any point $y \in U$, we have that $y \in A$ so that $y \notin X - A$, which shows that U does not intersect $X - A$. Since U is a neighborhood of x , this shows that $x \notin \overline{X - A}$ by Theorem 17.5 part (a). Then, since x was an arbitrary element of $\bar{A}$, it follows that $\bar{A}$ and $\overline{X - A}$ are disjoint so that $\text{Bd } A = \bar{A} \cap \overline{(X - A)} = \emptyset$ as desired.

(c) ($\Rightarrow$) First suppose that U is open. This implies that $X - U$ is closed and hence $X - U = \overline{X - U}$. Now by definition $\text{Bd } U = \bar{U} \cap \overline{(X - U)} = \bar{U} \cap (X - U) = \bar{U} - U$, as required.

($\Leftarrow$) Conversely, suppose that $\text{Bd } U = \bar{U} - U = \bar{U} \cap \overline{(X - U)}$. Given that $\bar{U} \cap (X - U) = \bar{U} - U$, the last equality amounts to

$$\bar{U} \cap \overline{(X - U)} = \bar{U} \cap (X - U) \quad (1)$$

which in turn implies that $\overline{(X - U)} = X - U$. This last equality holds because first $X - U \subseteq \overline{(X - U)}$ obviously. Conversely, given an element $x \in \overline{(X - U)}$, there are two possibilities: if $x \in \bar{U}$ then $x \in \bar{U} \cap \overline{(X - U)}$ and hence by the equality (1), $x \in \bar{U} \cap (X - U)$ and hence $x \in X - U$. Alternatively, if $x \notin \bar{U}$ then $x \notin U$ and hence $x \in X - U$ which completes the proof that $\overline{(X - U)} = X - U$ and proves that $X - U$ is closed. Then U is consequently open.

(d) We claim that this is not generally true. As a counterexample consider the set $U = \mathbb{R} - \{0\}$ as a subset of $\mathbb{R}$ (endowed with the standard topology). Clearly U is open as its complement is a singleton and singletons in Hausdorff spaces are always closed. It is also obvious that $\bar{U} = \mathbb{R}$. This implies that $\text{Int}(\bar{U}) = \text{Int}(\mathbb{R}) = \mathbb{R} \neq \mathbb{R} - \{0\} = U$.

Problem 7

Find the boundary and the interior of each of the following subsets of $\mathbb{R}^2$:

- (a) $A = \{(x, y) \mid y = 0\}$
- (b) $B = \{(x, y) \mid x > 0 \text{ and } y \neq 0\}$
- (c) $C = A \cup B$
- (d) $D = \{(x, y) \mid x \text{ is rational}\}$
- (e) $E = \{(x, y) \mid 0 < x^2 - y^2 \leq 1\}$
- (f) $F = \{(x, y) \mid x \neq 0 \text{ and } y \leq 1/x\}$

Solution: (a) It is easy to show that A is closed so that $\bar{A} = A$, and that also $\overline{\mathbb{R}^2 - A} = \mathbb{R}^2$ so that $\text{Bd } A = A$. It is also easy to see that no basis element and therefore no neighborhood of any point in A is contained entirely within A . From this it follows that $\text{Int } A = \emptyset$.

(b) It is easy to show that B is open so that $\text{Int } B = B$. It is likewise not difficult to prove that $\bar{B} = \{(x, y) \mid x \geq 0\}$. We then have from Exercise 17.19 part (c) that $\text{Bd } B = \bar{B} - B = \{(x, y) \mid x = 0\} \cup \{(x, y) \mid x > 0 \text{ and } y = 0\}$.

(c) Here we have that $C = A \cup B = \{(x, y) \mid y = 0\} \cup \{(x, y) \mid x > 0\}$. It is then easy to show that the closure is $\bar{C} = \{(x, y) \mid y = 0\} \cup \{(x, y) \mid x \geq 0\}$. We also have that $\mathbb{R} - C = \{(x, y) \mid x \leq 0 \text{ and } y \neq 0\}$ so that $\overline{\mathbb{R} - C} = \{(x, y) \mid x \leq 0\}$. From these we clearly then have

$$\text{Bd } C = \bar{C} \cap \overline{(\mathbb{R} - C)} = \{(x, y) \mid x < 0 \text{ and } y = 0\} \cup \{(x, y) \mid x = 0\}$$

It is also not difficult to show that $\text{Int } C = \{(x, y) \mid x > 0\}$.

(d) Clearly we have that $\bar{D}$ is all of $\mathbb{R}^2$ as a consequence of the fact that the rationals are order-dense in the reals. Also, since any neighborhood of any point in D will intersect a point (x, y) with irrational x , it follows that no point of D is in its interior. Thus $\text{Int } D = \emptyset$ so that $\bar{D} = \text{Int } D \cup \text{Bd } D = \emptyset \cup \text{Bd } D = \text{Bd } D$ by Exercise 17.19 part (a), and hence $\text{Bd } D = \bar{D} = \mathbb{R}^2$.

(e) It should be fairly obvious by this point that

$$\text{Bd } E = \{(x, y) \mid |y| = |x|\} \cup \{(x, y) \mid x^2 - y^2 = 1\}$$

(f) First we clearly have that $\text{Int } F = \{(x, y) \mid x \neq 0 \text{ and } y < 1/x\}$. We also have that $\bar{F} = \{(x, y) \mid x = 0\} \cup \{(x, y) \mid x \neq 0 \text{ and } y \leq 1/x\}$. By Exercise 17.19 part (a) we have that $\bar{F} = \text{Int } F \cup \text{Bd } F$ and that $\text{Int } F \cap \text{Bd } F = \emptyset$ so that $\text{Bd } F = \bar{F} - \text{Int } F$. Thus we have that $\text{Bd } F = \{(x, y) \mid x = 0\} \cup \{(x, y) \mid x \neq 0 \text{ and } y = 1/x\}$. Again these facts are not difficult to show rigorously but would be tedious.